

- 6 (a) Loss in EPE of the E- field = Gain in KE of the electron

$$qV = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2qV}{m}} = \sqrt{\frac{2(1.6 \times 10^{-19})(1000)}{(9.11 \times 10^{-31})}}$$

$$= 1.8742 \times 10^7 \text{ m s}^{-1} = 1.87 \times 10^7 \text{ m s}^{-1}$$

- (b) constant horizontal velocity as no horizontal accelerating forces (for vertical plates) (similar to projectile motion where horizontal velocity is usually constant).

$$\text{time taken} = \frac{6.0 \times 10^{-2}}{1.8742 \times 10^7}$$

$$= 3.20 \text{ ns (3 sf)}$$

$$= 3.2 \text{ ns (2 sf)}$$

Must show substitution and +1 sf to get M1
Allow e.c.f.

- (c) Actual p.d. across plates = $5.0 \times 20 = 100 \text{ V}$

$$\text{Electric field strength } E = \frac{V}{d} = \frac{100}{4.0 \times 10^{-2}} = 2500 \text{ N C}^{-1}$$

$$\text{Vertical acceleration} = \frac{qE}{m} = \frac{(1.6 \times 10^{-19})(2500)}{9.11 \times 10^{-31}} = 4.390770 \times 10^{14} \text{ m s}^{-2}$$

$$\text{Vertical velocity upon exiting vertical deflection plates}$$

$$= u + at = (4.39 \times 10^{14})(3.2 \times 10^{-9}) = 1.405 \times 10^6 \text{ m s}^{-1}$$

$$\text{Time taken for rest of } 14 \text{ cm} = \frac{14 \times 10^{-2}}{1.874 \times 10^7} = 7.470 \times 10^{-9} \text{ s}$$

Vertical displacement for first 6.0 cm

$$= \frac{1}{2}at^2 = \frac{1}{2}(4.390770 \times 10^{14})(3.2 \times 10^{-9})^2 = 2.248 \times 10^{-3} \text{ m}$$

Vertical displacement for next 14.0 cm

$$= ut = (1.405 \times 10^6)(7.470 \times 10^{-9}) = 1.0495 \times 10^{-2} \text{ m}$$

Distance between vertical divisions

$$= 1.0495 \times 10^{-2} + 2.248 \times 10^{-3} = 1.274 \times 10^{-2} \text{ m}$$

$$= 1.3 \times 10^{-2} \text{ m}$$

- (d)
- smaller amplitude / deflection
 - because same charge, same force, but larger mass (1000x), so deflection would be smaller

OR

- waveform is compressed horizontally
- because same charge, same force, but larger mass (1000x), so the applied voltage will not be sufficient to deflect the electron beam to the extreme ends.

OR

- waveform is inverted / reflected about the x-axis (timebase axis) and y-axis (voltage gain axis)
- because opposite charge, deflection will be in the opposite direction